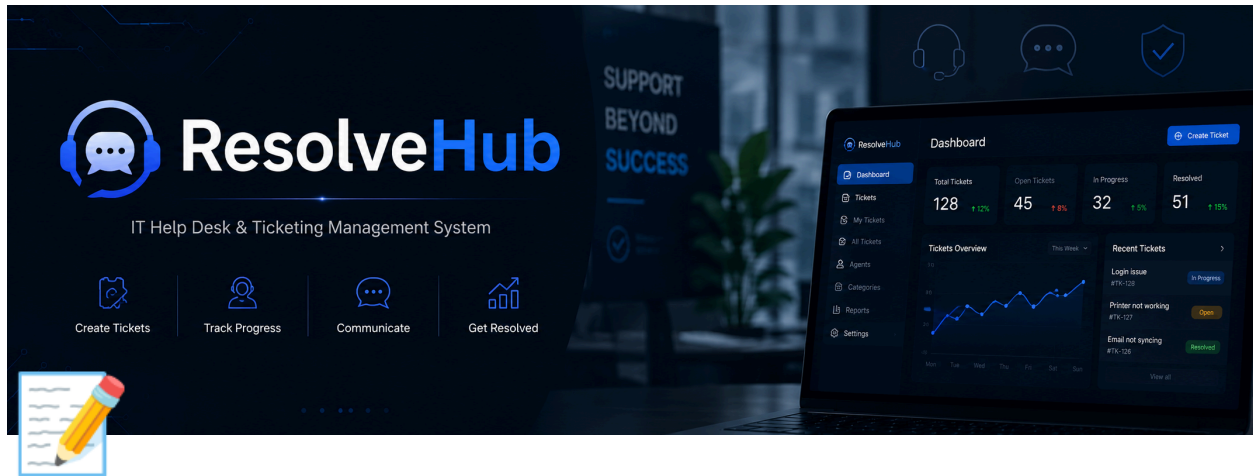




Requirement Analysis

This section defines the functional and non-functional requirements of ResolveHub. It outlines the system features, expected behavior, performance, security, usability, and other quality requirements that guide the design and development of the application.

Functional Requirements:

1. **Authentication and User Management:** ResolveHub shall provide a secure authentication system that allows users to access the platform based on their roles.

Requirements:

- Allow users to register and login using email and password.
- Encrypt user passwords before storing them in the database.
- Validate password strength.
- Support forget password and reset password functionality.
- Use JWT authentication for secure session management.
- Protect private pages and API routes.
- Apply role-based access control.

- Allow users to update their profile information.
- Log important user activities such as login, logout, ticket creation, and ticket updates.

Authentication protects the system from unauthorized access. Role-based authorization ensures that employees, agents, managers, and admins only access the features related to their responsibilities.

2. **Ticket Management:** The ticket management module is the core feature of ResolveHub. It allows employees to create support requests and allows the IT teams to manage those requests until they are resolved.

Requirements:

The system shall:

- Allow employees to create support tickets.
- Generate a unique ticket reference number for each ticket.
- Require important ticket fields such as title, description, category, and priority.
- Allow users to edit tickets before they are assigned or resolved.
- Allow users to cancel tickets if the issue is no longer needed.
- Store ticket creation date and update date.
- Support ticket search and filtering.
- Track the full history of ticket changes.

Ticket Categories:

- Hardware
- Software
- Network
- Email
- Access Request
- Security

- Other

Ticket priorities:

ResolveHub shall support the following priority levels:

- Low
- Medium
- High
- Critical

Ticket Statuses:

ResolveHub shall support the following statuses:

- Open
- Assigned
- In Progress
- Pending
- Resolved
- Closed
- Cancelled

3. **Ticket Assignment and Workflow:** The system shall support a structured workflow for assigning and resolving tickets.

Requirements:

The system shall:

- Allow admins to assign tickets to IT support agents.
- Allow ticket reassignment when needed.
- Support manual assignment by admin.
- Support automatic assignment.
- Allow agents to update the ticket status.
- Track assignment history.

- Store all ticket actions in an audit trail.
- Allow escalation for urgent or unresolved tickets.
- Allow internal comments for IT staff only.

Ticket Lifecycle:

The ticket lifecycle in ResolveHub follows this process:

Open → Assigned → In Progress → Pending → Resolved → Closed

4. **Communication and Notifications:** ResolveHub shall include a communication system inside each ticket to keep all discussions organized.

Requirements:

The system shall:

- Allow users to receive in-app notifications for important system events.
- Send email notifications for ticket creation, assignments, status updates, and resolution.
- Notify users whenever the status of the ticket is updated.
- Allow employees and IT support system to communicate through a comment and reply system within each ticket.
- Allow users to mention or tag IT support agents in ticket comments using the @username format.
- Display a notification center dashboard where the users can view all recent notifications.
- Update notifications in real time.

Instead of using WhatsApp messages, phone calls, or scattered emails, all communication will be saved inside the ticket. This makes the support process clear, professional and easy to review.

5. **Dashboard and Reporting:** ResolveHub shall provide dashboards that help each user role understand the most important information quickly.

Employee Dashboard:

The employee dashboard shall show:

- Total submitted tickets.
- Open tickets
- In-progress tickets.
- Resolved tickets
- Latest notifications

Agent Dashboard:

The agent dashboard shall show:

- Assigned tickets.
- Tickets in progress
- Pending tickets.
- High-priority tickets.
- Critical tickets.
- Resolved tickets.
- Recent comments or updates.

Admin Dashboard:

The admin dashboard shall show:

- Total number of tickets.
- Open tickets.
- Pending tickets.
- Resolved tickets.
- Closed tickets.
- Tickets by category.
- Tickets by priority.
- Agent workload.
- Recent system activity.
- Monthly ticket statistics.

- Agent Performance chart

Manager Dashboard:

The manager dashboard shall show:

- Team-related tickets.
 - Department ticket progress.
 - Common issue category.
 - Unresolved tickets.
 - Monthly reports.
 - Ticket priority distribution.
-

Reports:

ResolveHub shall include reporting features to help admins and managers analyze support performance.

Reports include:

- Monthly ticket reports
- Tickets by category.
- Tickets by priority.
- Average resolution time.
- Agent performance.
- Employee activity reports.
- Pending and delayed tickets.
- SLA reports.
- Charts and analytics.

Export Options:

The system should allow reports to be exported as:

- PDF
- Excel

6. **File Attachments:** ResolveHub shall allow users to upload files to support their tickets.

Requirements:

The system shall:

- Allow employees to upload screenshots, documents, or log files.
- Allow agents to download attachments securely.
- Validate file size before upload.
- Validate supported file types.
- Store file information in the database.
- Prevent unauthorized users from accessing private attachments.

Example: If an employee receives an error message, they can upload a screenshot so the IT agent can understand the problem faster.

7. **Admin Panel:** ResolveHub shall include an admin panel for managing the whole system.

Requirements:

The admin panel shall allow admins to:

- Manage users.
- Manage roles.
- Manage ticket categories.
- Manage priorities and statuses.
- View all tickets.
- Assign and reassign tickets.
- View reports.
- View activity logs.
- Manage system settings.
- Monitor system activity.

8. **AI-Powered Features:**

a. AI Ticket Categorization:

The system shall:

- Analyze the ticket title and description.
- Suggest the most suitable category for the ticket.
- Allow the employee or admin to accept or change the suggested category.

b. AI Priority Suggestion:

The system shall:

- Analyze the ticket description.
- Suggest a priority level: Low, Medium, High, or Critical.
- Allow the admin or agent to override the suggested priority.

c. AI Suggested Replies:

The system shall:

- Generate suggested troubleshooting replies for IT agents.
- Allow the agent to edit the AI-generated reply before sending it.
- Help agents respond faster to common issues.

d. AI Duplicate Ticket Detection:

The system shall:

- Detect tickets with similar descriptions.
- Recommend existing tickets before creating a new one.
- Help reduce duplicate support requests.

e. AI Ticket Summary:

The system shall:

- Generate AI-powered summaries of ticket conversations.
- Highlight important actions and resolutions.
- Allow IT support agents to review summaries before using them.

9. **Real-time Chat Support:**

The system shall:

- Allow employees and IT support agents to communicate through a real-time chat interface.
- Display typing indicators and online status.
- Store chat history for future reference.
- Notify users instantly when a new message is received.
- Allow file sharing within chat conversations.

10. **SLA Timer System:**

The system shall:

- Display SLA countdown timers.
- Track response and resolution deadlines for every ticket.
- Highlight tickets approaching their SLA deadline.
- Notify agents and administrators when an SLA is about to expire.
- Mark SLA violations in reports.

11. **Audit Logs:**

The system shall:

- Record all important user activities.
- Track login and logout events.
- Track ticket creation, assignment, reassignment, updates, and deletion.
- Record administrative actions.

12. **QR Code Asset Management:**

The system shall:

- Generate a unique QR code for each registered IT asset.
- Allow users to scan QR codes to view asset information.
- Link support tickets to scanned assets.

- Maintain the maintenance history of each asset.
-

Non-Functional Requirements:

Non-functional requirements describe the quality, security, and performance expectations of the system.

1. **Usability:** ResolveHub should have a clear, modern, and user-friendly interface.

The system should:

- Be easy to use for employees and IT staff.
- Use a clean SaaS-style design.
- Provide simple navigation through a sidebar.
- Use clear buttons, forms, and status labels.
- Show loading states when data is being processed.
- Show helpful error messages when something goes wrong.
- Support both desktop and mobile screen sizes.

2. **Responsiveness:** The system should be responsive and work smoothly on:

- Desktop screens
- Laptop screens
- Tablets
- Mobile phones

3. **Security:** ResolveHub should protect user accounts and company supports data.

The system should:

- Use secure login and authentication.
- Store passwords in encrypted form.
- Use JWT tokens for secure sessions.

- Restrict access based on user roles.
- Protect API routes.
- Validate user input.
- Prevent unauthorized access to tickets and attachments.
- Record important actions in activity logs.

4. **Performance:** The system should load quickly and handle support requests efficiently.

The system should:

- Load dashboards quickly.
- Return search and filter results quickly.
- Handle multiple users at the same time.
- Store and retrieve ticket data efficiently.
- Avoid unnecessary API requests.

5. **Reliability:** ResolveHub should be reliable and preserve important ticket data.

The system should:

- Save tickets correctly.
- Keep ticket history even after updates.
- Prevent accidental loss of comments or attachments.
- Show clear error messages if an operation fails.
- Keep audit logs for important actions.

6. **Maintainability:** The project should be organized in a clean way so it can be improved later.

The system should:

- Separate frontend, backend, and database logic.
- Use reusable UI components.
- Use clear API structure.

- Use organized folders and meaningful file names.
- Include documentation for setup and usage.
- Use GitHub for version control.

7. **Scalability:** ResolveHub should be designed so it can grow in the future.

The system should allow future additions such as:

- More departments.
- More user roles.
- More ticket categories.
- Real-time chat.
- SLA tracking.
- AI automation.
- Multi-language support.
- Mobile application version.